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December 2023

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Air quality and employee performance in teams: Evidence from the NFL. Economics and Human Biology, 51: 101310, 2023.

Anterior cruciate ligament injury occurrence, return to sport and subsequent injury in the Australian High Performance Sports System: A 5-year retrospective analysis. Physical Therapy in Sport, 65: 140-146, 2023.

Cannabis use in adolescent and young adult athletes: A clinical review. Sports Health, Ahead of Print: 1-6, 2023.

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Epidemiologic trends in hand injuries in the National Football League from 2009-2010 to 2019-2020. The Physician and Sportsmedicine, Ahead of Print: 1-26, 2023.

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Epidemiology of quadriceps muscle strain injuries in elite male Australian football players. Scandinavian Journal of Medicine & Science in Sports, Ahead of Print: 1-11, 2023.

Examining whether loss of consciousness is associated with worse performance on the SCAT5 and slower clinical recovery after concussion in professional athletes. Journal of Neurotrauma, 40: 2330-2340, 2023.

Exploring the role of sprint biomechanics in hamstring strain injuries: A current opinion on existing concepts and evidence. Sports Medicine, Ahead of Print: 1-11, 2023.

Managing lower limb muscle reinjuries in athletes: From risk factors to return-to-play strategies. Journal of Functional Morphology and Kinesiology, 8: 155, 2023.

Nutritional priorities, practices and preferences of athletes and active individuals in the context of new product development in the sports nutrition sector. Frontiers in Sports and Active Living, 5: 1088979, 2023.

Performance Management in Elite Football: A teamwork modeling approach. International Journal of Sports Physiology and Performance, Ahead of Print: 1-3, 2023.

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Relationship between repeated sprint ability, countermovement jump and thermography in elite football players. Sensors, 23: 631, 2023.

Editorial Commentary: Return to sports, return to preinjury sports, return to prior performance at sports: All meaningful parameters with different meanings and scales. Sports Health. Ahead of Print: 1-3, 2023.

Substance misuse in elite athletes: Early detection, brief intervention and referral to treatment. Clinics in Sports Medicine, Ahead of Print: 1-18, 2023.

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The association between hip/groin pain and hip strength in football players: An exploratory analysis of the FORCe cohort. Journal of Science and Medicine in Sport, Ahead of Print: 1-25, 2023.

The role of strength-related factors on psychological readiness for return to sport following anterior cruciate ligament (ACL) reconstruction. Healthcare, 11: 2787, 2023.

Utilizing extracorporeal shockwave therapy for in-season athletes. Healthcare, 11: 1006, 2023.

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A brief review of the literature for published dual-energy x-ray absorptiometry protocols for athletes. Sports Health, Ahead of Print: 1-9, 2023.

Context: Dual-energy x-ray absorptiometry (DXA) is widely known for its utility in diagnosing a patient with osteopenia or osteoporosis; however, its utility in evaluation of body composition and potential athletic performance was previously routinely overlooked. In recent years, athletic programs have begun employing this equipment during athlete screening. However, it is currently unknown how athletic programs are utilizing this information to guide an athlete's training and health.

Objective: To explore the literature to identify what is known regarding sports performance and athlete health based on body composition recordings.

Data Sources: PubMed and Google Scholar databases were searched for this brief review.

Study Selection: A focus was placed on articles within the past 10 years that discussed DXA protocols within athletic populations; 14 articles were included in this brief literature review.

Study Design: Brief literature review.

Level of Evidence: Level 5.

Data Extraction: One member of the research team searched the literature and retrieved articles with the purpose of analyzing and/or explaining DXA imaging in body composition analysis of active persons (primarily athletes).

Results: Quality assurance scans with a phantom calibration block as well as athlete prescreening condition and activity standardization was routinely recommended. However, only 1 study reported a specific DXA protocol for athletes, and only 1 study described guidelines for how to report DXA results in athletic populations, suggesting it is plausible yet difficult due to the small changes detectable.

Conclusion: Due to the limited literature as well as a lack of reference values for specific athletic populations, the authors of this review recommend using the current Nana et al (Int J Sport Nutr Exerc Metab 2015;25:198-215) DXA protocol for performing DXA scans in the athletic population as well as current Hind et al (J Clin Densitom 2018;21:429-443) guidelines for distributing the information.

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A potential role exists for nutritional interventions in the chronic phase of mild traumatic brain injury, concussion and sports-related concussion: A systematic review. Nutrients, 15: 3726, 2023.

Abstract - Mild traumatic brain injury (mTBI) represents a significant burden for individuals, economies, and healthcare systems worldwide. Recovery protocols focus on medication and physiotherapy-based interventions. Animal studies have shown that antioxidants, branched-chain amino acids and omega-3 fatty acids may improve neurophysiological outcomes after TBI. However, there appears to be a paucity of nutritional interventions in humans with chronic (≥1 month) symptomology post-mTBI. This systematic literature review aimed to consolidate evidence for nutrition and dietary related interventions in humans with chronic mTBI. The review was registered with the International Prospective Register of Systematic Reviews (PROSPERO; CRD42021277780) and conducted following the Preferred Reporting for Systematic Reviews and Meta-Analyses (PRISMA) guidelines. Three reviewers searched five databases (PubMed/MEDLINE, Web of Science, SPORTDiscus, CINAHL Complete and Cochrane), which yielded 6164 studies. Nine studies met the inclusion criteria. The main finding was the lack of interventions conducted to date, and a quality assessment of the included studies was found to be fair to good. Due to heterogeneity, a meta-analysis was not feasible. The six nutrition areas identified (omega-3 fatty acids, melatonin, Enzogenol®, MLC901, ketogenic diet and phytocannabinoids) were safe and well-tolerated. It was found that these nutritional interventions may improve cognitive failures, sleep disturbances, anxiety, physical disability, systolic blood pressure volume and sport concussion assessment tool scores following mTBI. Potential areas of improvement identified for future studies included blinding, reporting compliance, and controlling for confounders. In conclusion, further research of higher quality is needed to investigate the role of nutrition in recovery from mTBI to reduce the burden of chronic outcomes following mTBI.

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Air quality and employee performance in teams: Evidence from the NFL. Economics and Human Biology, 51: 101310, 2023.

Abstract - Emerging research documents the impact of poor air quality on employee performance in a number of settings, including sport. Unlike other settings, sports teams have limited ability to influence previously scheduled games, making air quality exogenous to unobservable factors affecting game outcomes. We link play-level data from National Football League (NFL) games to data from air quality monitoring stations located near NFL stadiums. Results using data from about 90,000 offensive plays in 900 NFL games indicate that yards gained per offensive play increase with $PM_{2.5}$ concentration. Defensive players are relatively more affected by air quality than offensive players.

Anterior cruciate ligament injury occurrence, return to sport and subsequent injury in the Australian High Performance Sports System: A 5-year retrospective analysis. Physical Therapy in Sport, 65: 140-146, 2023.

Objectives: To report anterior cruciate ligament (ACL) injury occurrence, return to sport (RTS) timeframes and ACL subsequent injuries recorded in the Australian High Performance Sports System according to athlete sex.

Methods: ACL injury data of injured athletes were prospectively collected by the treating sports and exercise physicians and physiotherapists within the Australian High Performance Sports System between January 1, 2015 and May 31, 2020. RTS time frames for ACL injuries and the proportion of subsequent ACL injuries were calculated. The RTS time was compared between sexes, age groups and ACL injury categories using the Mann-Whitney U test and the Kruskal-Wallis test.

Results: A total of 132 ACL injuries were reported in 108 athletes (77 female, 31 male). ACL injuries accounted for 6.4% of all reported knee injuries, with almost half (48.5%) of the ACL injuries reported to occur during training. Median RTS time was 369 days (IQR = 273–487), with RTS times reducing with increasing age (X2 (Zbrojkiewicz et al., 2018) = 11.781, p = 0.008). The presence of concurrent knee pathology did not significantly affect the RTS timeframes. ACL injuries were most frequently reported in netball, winter sports, basketball, field hockey and gymnastics. One quarter of the ACL reported (n = 34,

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25.8%) were subsequent to a prior ACL injury, with the majority of these injuries occurring to the ipsilateral knee (n = 26) as opposed to contralateral knee (n = 6).

Conclusions: Despite ACL injuries accounting for a small proportion of all knee injuries reported in the Australian High Performance Sports System, more than a quarter are subsequent to a previous ACL injury. Shorter RTS times were observed in older athletes; however, considerable periods of time-loss occur as a consequence of ACL injuries. Effective primary prevention, rehabilitation processes, and ongoing tertiary prevention strategies are warranted in this population.

Cannabis use in adolescent and young adult athletes: A clinical review. Sports Health, Ahead of Print: 1-6, 2023.

Context: Cannabis use among the general population has increased over time, in part due to decriminalization of use and greater social acceptance of cannabis use. These changes have contributed to increased availability of cannabis products, thus raising the likelihood that a subset of adolescent and young adult athletes will use cannabis. Therefore, it is imperative that clinicians and other providers working with young athletes have a thorough understanding of the impact cannabis can have on the athletic performance and overall health of a young athlete.

Evidence Acquisition: PubMed (Medline), CINAHL, PsycINFO, and SportDiscus databases were used to perform a literature search of articles published between January 1, 2010 and December 31, 2022. Additional articles were reviewed based on references obtained from initial articles.

Study Design: Clinical review.

Level of Evidence: Level 4.

Results: Survey studies suggest that up to 1 in 4 athletes have used cannabis at least once in the last year. Age, sex, race, sexual orientation, level of competition, and country of residence of an athlete all contribute to differing rates of cannabis use among athletes. The scientific literature does not support using cannabis for athletic performance, and multiple studies have demonstrated notable impairments in objective athletic performance measures. Cannabis use can also negatively impact an athlete's overall health via cardiovascular, gastrointestinal, and mental health symptoms and disorders.

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Conclusion: Cannabis use among adolescent and young adult athletes is common, and rates of use are influenced by many different factors. Current evidence suggests that cannabis use can worsen sport performance, negatively impact an athlete's general health, and contribute to negative mental health outcomes.

Characteristics and outcomes of athletes with slow recovery from sports-related concussion: A CARE Consortium Study. Neurology, Vol. 100(14): e1510-e1519, 2023.

Background and Objectives: Some athletes experience a slow recovery after sport-related concussion (SRC). There is little agreement on what constitutes slow recovery, however, and minimal data on the prevalence, predictors, or prognosis for this group. The objectives of this study were to apply an operationalized definition of slow recovery and characterize predictors and long-term prognosis of these individuals.

Methods: This is a prospective multisite observational study of collegiate athletes. Participants underwent multimodal assessments preseason and 5 additional time points after SRC. Time from injury to initiation of return to play progression (asymptomatic timepoint) and from injury to return to play (RTP) were the primary markers of recovery.

Results: One thousand seven hundred fifty-one concussed male and female collegiate athletes were studied. Eighty percent of participants reached the asymptomatic and/or RTP time points by days 14 and 24, respectively. Slow recovery was thus defined as exceeding 1 or both of those intervals (n = 399). This group was statistically more likely to be female (41.1% vs 35.6%, $p = 0.05$), have higher initial postinjury SCAT symptom severity scores (mean [SD]: 36.6 [23.4] vs 25.4 [19.9], $p < 0.001$), lower postinjury Standardized Assessment of Concussion scores (mean [SD]:25.74 [2.98] vs 26.26 [2.85], $p = 0.004$), perform worse on the postinjury Balance Error Scoring System (mean [SD]: 17.8 [8.9] vs 15.9 [8.5], $p < 0.01$), have fewer assessments in the first 14 days after injury (mean [SD]: 48.8 [29.7] vs 67.9 [24.6], $p < 0.01$), and be injured in practice (70.7% vs 65.1%, $p = 0.04$). 77.6% of the slow recovery group returned to play within 60 days of injury, and 83.4% (n = 349) returned to play within 90 days of injury. Only 10.6% had not returned to play 6 months postinjury.

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Discussion: This study suggests an overall favorable prognosis for slowly recovering athletes and provides data for athletes and medical teams to consider in calibrating RTP expectations and making decisions about medical disqualification vs ongoing engagement in their sport.

Chronic cannabis consumption and physical exercise performance in healthy adults: A systematic review. Journal of Cannabis Research, 2:34, 2023.

Objective: The effects of chronic cannabis consumption on physiological parameters of athletic performance are investigated to determine whether chronic cannabis consumption negatively affects athletic performance; improves performance, potentially via enhanced recovery; or has no effect at all.

Methods: A systematic review of the literature (cross-sectional, longitudinal, and intervention studies) concerning the effects of cannabis consumption on sports performance outcomes, e.g. VO2Max (maximal oxygen uptake), PWC (physical work capacity) up to January 2020 was conducted using the PubMed, CINAHL, Medline, PsycArticles, PsycInfo, SPORTDiscus, Psychology and Behavioural Sciences Collection, and Health Source: Nursing/Academic Edition databases. After screening and additional forward searching, four articles were found to fit the inclusion criteria.

Results: Resting heart rate was the only physiological measure that significantly differed between groups, and only in one of the four studies included herein. The strongest predictors of athletic performance (VO2Max and PWC) were not found to be significantly different between groups in any of the included studies. Chronic cannabis consumption had no significant effect on athletic performance. The included studies did not assess other elements, such as recovery or endurance.

Conclusion: No evidence exists for ergogenic or ergolytic effects from chronic cannabis consumption. In some sports, advantages may plausibly be conveyed by psychotropic enhancement or pain reduction. Further research (particularly longitudinal or interventional studies) is required to determine whether cannabis, or constituents thereof, may provide indirect supplemental benefits to athletes.

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Early tendon morphology as a biomarker of long-term patient outcomes after surgical repair of Achilles tendon rupture – A prospective cohort study. The Orthopaedic Journal of Sports Medicine, 11(11): 1-8, 2023.

Background: Patient outcome after acute Achilles tendon rupture (ATR) varies and is difficult to predict. Whether early variations in healing, visualized with ultrasonography, can predict long-term patient outcome is unclear.

Purpose: To (1) examine the associations of Achilles tendon cross-sectional area (CSA) and elongation (TE) during healing of ATR repair with patient outcomes at 12 months postoperatively and (2) investigate the predictive or diagnostic capacity of the morphological biomarkers.

Study Design: Cohort study; Level of evidence, 2.

Methods: This study was based on previously collected data from 86 patients who underwent acute standardized ATR repair between 2013 and 2018 and who were included in a prior randomized trial investigating early functional mobilization (EFM). In the EFM group, loading was allowed immediately after surgery, while in the comparison group, loading was allowed first at 2 weeks postoperatively. Achilles tendon CSA and length were measured with ultrasound at 6 weeks, 6 months, and 12 months postoperatively. CSA ratio and absolute difference in the length of the healthy and injured tendons were calculated. Patient-reported outcome was registered with the validated Achilles tendon Total Rupture Score and functional outcome with the heel-rise endurance test at 12 months postoperatively. The limb symmetry index (LSI) was calculated for maximum heel-rise height (HRHmax) and total concentric work. Multiple linear regression adjusted for age was used, and the area under the receiver operating characteristic curve (AUC) was calculated to evaluate predictive capacity.

Results: A larger CSA ratio at 6 weeks was associated with higher LSI HRHmax at 12 months (R^2, 0.35; $P < .001$) and exhibited good predictive capacity (AUC, 0.82). More TE at 12 months was associated with lower LSI total concentric work at 12 months (R^2, 0.21; $P = .001$) and exhibited acceptable predictive capacity (AUC, 0.71).

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Conclusion: Greater Achilles tendon CSA seen on ultrasound 6 weeks after surgical repair had good clinical prediction for long-term functional outcome. TE at 12 months was predictive of inferior functional outcome.

Epidemiologic trends in hand injuries in the National Football League from 2009-2010 to 2019-2020. The Physician and Sportsmedicine, Ahead of Print: 1-26, 2023.

Objectives: In American football, hand injuries have been shown to negatively impact performance. The purpose of this study is to characterize the prevalence and severity of hand injuries in National Football League (NFL) players.

Methods: A public online database was utilized to identify hand injuries in NFL players from 2009–2010 to 2019–2020. The primary outcome was to analyze the overall incidence of hand injuries (including wrist, metacarpus, finger, and thumb), injury type by each aforementioned anatomic location, and player position. Injury severity was evaluated based on percentage of injuries in which players returned to play (RTP), number of games missed before RTP, and the percentage of injuries resulting in the player being placed on injured reserve (IR).

Results: Of the 6,127 players included, 847 (13.8%) players sustained a hand injury, of which 24.8%, 34.3%, 17.9%, and 22.9% occurred at the wrist, metacarpus, finger, and thumb, respectively. Of the injured players, 97.4% returned to play following their injury, 14.8% were put on IR, and an average of 1.7 (SD 3.3) games were missed. Quarterbacks were the most likely to sustain hand injuries at all anatomic locations. Wrist injuries were associated with the lowest RTP rate (93.3%), the most players placed on injured reserve (28.6%), and the greatest number of games missed (mean 2.5, SD 4.2).

Conclusion: Hand injuries decreased in prevalence by 65.6% over the 11 NFL seasons evaluated. This trend coincides with the implementation of several safety rules that relate to components of play involving the hands. Quarterbacks experienced the greatest prevalence and severity for all hand injuries. Wrist injuries represent the anatomic location associated with the greatest severity. These findings may

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be able to inform tailored injury prevention practices by position, and advocate for the further adoption of safety rules to protect players from further injury.

Epidemiology of quadriceps muscle strain injuries in elite male Australian football players. Scandinavian Journal of Medicine & Science in Sports, Ahead of Print: 1-11, 2023.

Objective: To describe the epidemiology of quadriceps muscle strain injury (QMSI) in elite Australian Football League (AFL) players, explore recovery milestones and determine whether recovery is impacted by factors such as injury type (index vs. re-injury), the primary muscle injured and the mechanism of injury.

Measures: All QMSI data reported to the Soft Tissue Injury Registry of the AFL from the 2014 to 2020 seasons were evaluated. Player demographic data, circumstances of injury, MRI reports and recovery outcomes following injury were extracted. Descriptive statistics and frequency distributions are presented. Recovery outcomes for injury type, primary muscle injured and the mechanism of injury were compared using univariate analyses.

Results: There were 164 QMSIs from 122 players reported (134 index; 30 re-in juries). Almost all (91.3%) QMSIs involved the rectus femoris. Half (48.4%) of the QMSIs occurred during kicking and most commonly affected the dominant kicking leg (72%). The majority occurred at training (64.6%). All re-injuries involved the rectus femoris, most occurred from kicking (63.0%) and within 6months of the preceding injury (70%). The mean return to play (RTP) time was 25.4days (95%CI=22.6–28.2) and rectus femoris injuries took around 14days longer to RTP than vastii injuries (p=0.001). QMSIs with a kicking mechanism took the longest to RTP of all injury mechanisms.

Conclusion: In AFL players, QMSIs occur mostly in the dominant leg from a kicking mechanism. Rectus femoris injuries are more prevalent and result in longer RTP time frames. Re-injuries exclusively involved the rectus femoris, primarily from kicking.

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Examining whether loss of consciousness is associated with worse performance on the SCAT5 and slower clinical recovery after concussion in professional athletes. Journal of Neurotrauma, 40: 2330-2340, 2023.

Abstract - Video surveillance has almost universally been employed by professional sports to identify signs of concussion during competition. This study examined associations between video-identified possible loss of consciousness (LOC), acute concussion evaluation findings, and recovery time in concussed professional rugby league players. Medical personnel and sideline video operators identified head impact events sustained during three seasons of National Rugby League (NRL) matches to determine the need for further medical evaluation. If a concussion was suspected, players were removed from play and underwent a Head Injury Assessment, including the Sports Concussion Assessment Tool, Fifth Edition (SCAT5). Video footage was later examined to identify signs of possible LOC (i.e., observed LOC, no protective action in the fall, and unresponsiveness or lying motionless). Possible LOC was identified in 99 of the 1706 head impact events (5.8%). The median duration of apparent unresponsiveness was 4.2 sec (M = 7.4, standard deviation [SD] = 12.8, interquartile range [IQR] = 2.5–6.6). In the 661 athletes for whom SCAT5 data were available, those with possible LOC endorsed more SCAT5 symptoms and performed worse on Maddocks questions, Standardized Assessment of Concussion (SAC) total scores, orientation, immediate recall, concentration, and delayed recall. For the 255 players with medically diagnosed concussions, SCAT5 data were available for 245. Concussed players with possible LOC performed significantly worse on Maddocks questions. How ever, there were no group differences in SCAT5 symptom endorsement, SAC total scores, orientation, immediate recall, concentration, delayed recall, or the modified version of the Balance Error Scoring System (mBESS) total errors. Further, the presence or absence of possible LOC was not associated with number of games missed or time to medical clearance for match play. The duration of possible LOC was not associated with the number of games missed or time to medical clearance for match play. According to video review in NRL players, brief LOC might be more common than previously thought. The present study reveals possible LOC is not predictive of missed games or time to recover following concussion.

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Exploring the role of sprint biomechanics in hamstring strain injuries: A current opinion on existing concepts and evidence. Sports Medicine, Ahead of Print: 1-11, 2023.

Abstract - Hamstring strain injuries are one of the most common injuries in sprint-based sports with the mechanism of injury considered the result of an interaction between applied mechanical strain and the capacity of the muscle to tolerate strain. To date, injury prevention and rehabilitation strategies have frequently focused on enhancing the capacity of the hamstrings to tolerate strain, with little consideration of factors directly influencing mechanical strain. Sprint running biomechanics are one factor proposed to influence the mechanical strain applied to the hamstrings that may be modified (towards reduced strain) within rehabilitation and injury prevention programs. This article aims to explore the theoretical mechanistic link between sprint running mechanics and hamstring strain injury, along with the available supporting evidence. In doing so, it hopes to provide practitioners with an understanding of mechanical parameters that may influence hamstring strain injury whilst also identifying areas for further research exploration.

Key Points

- Sprint running mechanics are proposed to, and widely believed to influence mechanical strain placed on the hamstrings.
- Several articles presented in this review provide evidence to support the mechanistic link between running mechanics and hamstring strain.
- Targeting mechanical factors influencing hamstring strain may be beneficial in the prevention and rehabilitation of hamstring strain injuries and should be considered as an area of further study.

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Managing lower limb muscle reinjuries in athletes: From risk factors to return-to-play strategies. Journal of Functional Morphology and Kinesiology, 8: 155, 2023.

Abstract - Muscle injuries and subsequent reinjuries significantly impact athletes, especially in football. These injuries lead to time loss, performance impairment, and long-term health concerns. This review aims to provide a comprehensive overview of the current understanding of muscle reinjuries, delving into their epidemiology, risk factors, clinical management, and prevention strategies. Despite advancements in rehabilitation programs and return-to-play criteria, reinjury rates remain alarmingly high. Age and previous muscle injuries are nonmodifiable risk factors contributing to a high reinjury rate. Clinical management, which involves accurate diagnosis, individualized rehabilitation plans, and the establishment of return-to-training and return-to-play criteria, plays a pivotal role during the sports season. Eccentric exercises, optimal loading, and training load monitoring are key elements in preventing reinjuries. The potential of artificial intelligence (AI) in predicting and preventing reinjuries offers a promising avenue, emphasizing the need for a multidisciplinary approach to managing these injuries. While current strategies offer some mitigation, there is a pressing need for innovative solutions, possibly leveraging AI, to reduce the incidence of muscle reinjuries in football players. Future research should focus on this direction, aiming to enhance athletes' well-being and performance.

Nutritional priorities, practices and preferences of athletes and active individuals in the context of new product development in the sports nutrition sector. Frontiers in Sports and Active Living, 5: 1088979, 2023.

Introduction: Sports nutrition is a rapidly growing sector with increasing demand for evidence-based nutritional products to support competitive and healthy lifestyles. The product development process for novel foods should rely heavily on end-user engagement to facilitate future success, however there is a dearth of published information available. An understanding of the practices and self-reported nutritional priorities of athletes and active individuals is required for the development of new food products, facilitating evidence-based product formulation.

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Methods: Participants were at least 18 years of age, actively participating in competitive sport or structured physical activity on at least two occasions per week. Participants were asked to undertake a comprehensive online survey assessing their nutritional practice, perceived nutritional priorities and preferences for product characteristics. Questions were developed on the basis of critical evaluation of the current scientific literature and the hosting of two scoping focus group sessions with prospective end-users.

Results: 405 individuals (29 ± 9 years) completed this questionnaire. 295 participants reported active participation in competitive sport while the remaining 110 participants undertook structured physical activity exclusively. When asked to rank their top three most sought-after product claims in sports nutrition, “enhanced muscular recovery” was the most prioritized receiving 101 first choice preferences (25%) and 295 top 3 preferences. Fifty-eight percent of participants reported taking nutritional supplements. Caffeine containing functional foods (excluding caffeine supplements) were the most commonly used functional food group. A very low incidence of functional food usage was reported otherwise. When asked to rank the importance of various food product attributes, “nutritional profile” was ranked as the most important with rating of 3.37 ± 0.7 out of 4 followed by “taste” and “accessibility”. Whole food nutritional products received the most first preference selections and most top 3 selections when presented with a number of popular performance and recovery products on the market.

Conclusions: The transition towards a food first approach in sports nutrition is vital for athletes and active individuals to achieve their goals; with the development of evidence-based functional foods, particularly with a focus on muscle recovery, endurance, and strength enhancement at the forefront for new food product design and innovation.

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Performance Management in Elite Football: A teamwork modeling approach. International Journal of Sports Physiology and Performance, Ahead of Print: 1-3, 2023.

Overview - Winning matches and getting 3 competition points is commonly established as a key performance indicator (KPI) for elite football (soccer) teams to achieve their objectives during the season. While winning is the responsibility of the football players, sport-science and -medicine practitioners' responsibility is to support them by increasing their availability for matches (a key contributing factor to winning the games and ensuring team success). Unfortunately, it is not uncommon to observe sport-science (eg, strength and conditioning coach, physiologist) and -medicine (e.g., physician, physiotherapist) practitioners working in silos in a football setting. As such, we are proposing a teamwork modeling approach for practitioners with elite football teams. Our model comprises specific KPIs to enhance players' performance while reducing injury risk. This model has the potential to attract the attention of practitioners and researchers aiming to prevent injuries and improving players' recovery for enhanced football performance.

Pulsed red and blue photobiomodulation for the treatment of thigh contusions and soft tissue injury: A randomized controlled trial. Journal of Sport Rehabilitation, Ahead of Print: 1-7, 2023.

Context: Contusion and soft tissue injuries are common in sports. Photobiomodultion, light and laser therapy, is an effective aid to increase healing rates and improve function after various injury mechanisms. However, it is unclear how well photobiomodulation improves function after a contusion soft tissue injury. This study aimed to determine the effects of a pulsed red and blue photobiomodulation light patch on muscle function following a human thigh contusion injury.

Design: Single-blinded randomized control trial design.

Methods: We enrolled 46 healthy participants. Participants completed 5 visits on consecutive days. On the first visit, participants completed a baseline isokinetic quadriceps strength testing protocol at 60°/s

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and 180°/s. On the second visit, participants were struck in the rectus femoris of the anterior thigh with a tennis ball from a serving machine. Immediately following, participants were treated for 30 minutes with an active or placebo photobiomodulation patch (CareWear light patch system, CareWear Corp). Following the treatment, participants completed the same isokinetic quadriceps strength testing protocol. Participants completed the treatment and isokinetic quadriceps strength test during the following daily visits. We normalized the data by calculating the percent change from baseline. We used a mixed model analysis of covariance, with sex as a covariate, to determine the difference between treatment groups throughout the acute recovery process.

Results: We found the active photobiomodulation treatment significantly increased over the placebo group, quadriceps peak torque during the 180°/s test (P = .030), and average power during both the 60°/s (P = .041) and 180°/s (P ≤ .001) assessments. The mean peak torque and average power of 180°/s, at day 4, exceeded the baseline levels by 8.9% and 16.8%, respectively.

Conclusions: The red and blue photobiomodulation light patch improved muscle strength and power during the acute healing phase of a human thigh contusion injury model.

Relationship between repeated sprint ability, countermovement jump and thermography in elite football players. Sensors, 23: 631, 2023.

Abstract - Football is a very demanding sport which requires players to exert maximum effort, producing fatigue and eventually injuries. Thermography can be used to detect fatigue and prevent its consequences through thermal asymmetries in the bilateral body areas; however, its adequacy for elite footballers has not been widely studied. Therefore, the objective of the present investigation was to determine the suitability of thermography to detect fatigue in male football players. For this reason, twenty participants were gathered into a pair of subgroups (low [0.05) were found in any of the RSA test variables between low and high thermal asymmetry groups for thighs and calves. On the other hand, the low thermal asymmetry hamstring group reported a smaller percentage difference in sprints for the first sprint (%Diff) and a larger percentage difference in sprints two and three with respect to the best sprint (%Best). For CMJs, the low thermal asymmetry hamstring group reported significantly higher values post-RSA test, indicating better performance. Accordingly, thermography can provide information

about performance in CMJ and RSA tests through hamstring asymmetries over 0.2 ◦C. Meanwhile, larger asymmetries than 0.2 ◦C in calves and thighs do not seem to be related to performance in these tests; therefore, coaches should consider if it is optimal to align players with high hamstring asymmetries.

Editorial Commentary: Return to sports, return to preinjury sports, return to prior performance at sports: All meaningful parameters with different meanings and scales. Sports Health. Ahead of Print: 1-3, 2023.

The most important question on an athlete's mind with a serious knee injury (besides asking about treatment and surgery) is "When will I be able to return to sports?" Fortunately, there has been a total resetting in the past 10 years of the treatment algorithms to accomplish a safe return to sports (RTS) and decrease the risk for a repeat knee injury. Previous anterior cruciate ligament (ACL) publications seldom listed RTS objective criteria, and "time from injury" was the usual nonscientific rule.[2,9] Today, we have many available assets to utilize for our athletes in defining the process to be followed for RTS with assistance from our physical therapy and athletic trainer colleagues. This is the team approach we follow that is so necessary for treatment of our injured athletes.[9]

The study by Svantesson et al[12] entitled "Only 10% of patients with a concomitant medial collateral ligament (MCL) injury return to their preinjury level of sport 1 year after ACL reconstruction: a matched comparison with isolated ACL reconstruction" reports their findings on RTS parameters. The authors employed in their study design necessary postoperative parameters recommended for clinical practice and inclusion in RTS publications. The authors used common patient-reported outcomes, the Knee Injury and Osteoarthritis Outcome Score, Patient Acceptable Symptom State, the Swedish Knee Self Efficacy Scale, and objective criteria with muscle-strength isokinetic concentric tests and muscle function tests (single-leg hop for distance, side hops timed, vertical hop). These later function tests are now viewed as a "standard of care" in RTS evaluations as a part of an objective battery of RTS tests. It is also recognized that research is necessary on the objective function tests selected for RTS evaluations with the idea of incorporating agility testing and more reactive athletic tasks that simulate athletic reactive activities.[9,15]

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Of special note is the incorporation of the ACL-Return to Sport after injury (ACL-RSI) assessment.[14] This psychological evaluation consists of 12 questions in categories of emotions, confidence, and risk appraisal (0 to 10 Likert scale), with the highest value representing complete and total confidence for returning to athletics. From a reality standpoint, after an ACL injury and surgery, the question is to determine the loss of confidence with low values on the ACL-RSI scales. The realization is that our younger athletes (and even older athletes) have serious fears of reinjury and loss of confidence that require our attention and further evaluation before the return to athletics.[14]

The development of the Swedish National Knee Ligament Registry, comprising 90% of all ACL reconstructions in Sweden, and the Project ACL registry are major advances for clinical research and the founders and researchers are to be congratulated. Resulting registry publications have often strengthened treatment approaches and led to major changes in treatment by detecting failures, complications, and overall negative or successful outcomes.

In the Svantesson et al[12] study, the 1-year follow-up of ACL reconstructions (with and without an associated MCL injury) reported only 10.0% and 25.6%, respectively, return to their preinjury activity levels. This may initially appear to be a negative final result until further analysis is performed. It is true that publications showing an initial lower return to preinjury athletics lets our patients know that, even with expert surgery and rehabilitation, there are good reasons for delaying RTS past 9 to 12 months after ACL reconstruction and especially with concomitant injuries. Studies have shown that at 6 and 9 months after ACL reconstruction, the majority of athletes do not pass knee function and strength testing. Further, early RTS may lead to a 4 times or higher reinjury rate.[3,5,9] In addition to passing RTS tests, there is the added requirement for female athletes to undergo neuromuscular training to decrease the risk of a repeat ACL injury.9 For over 2 decades, we have recommended the nonprofit SportsMetrics neuromuscular training program that has proven statistically significant in decreasing ACL injuries and increasing athletic performance in female athletes.[8]

In the Svantesson et al[12] study, the reporting of the low preinjury sports at 1 year, along with the function and strength tests having returned to normal levels, raises the need for further questions. Granted there were only 30 patients in the ACL-MCL group; however, the return of only 3 of 30 ACL patients needs explanation. The question arises whether, given the 10% or even 25% return to preinjury sports, data in the ACL-MCL or ACL groups are truly representative of the final ACL outcomes. The study did show an overall RTS at a Tegner score ≥6 in 46.7% and 48.9% of the ACL-MCL and ACL groups,

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respectively, which is a more positive outcome. Of importance, the patient demographics shows approximately 60% of the athletes' preinjury sports were at very high Tegner athletic levels of 8 to 10. This corresponds to highly competitive and national elite athletes and helps us understand there was a shift from this highest athletic rating downward to participation in recreational sports. Perhaps the title should have the addition of "highly competitive athletes" in the title instead of "patients."

It would have been ideal in the follow-up analysis to pose a straightforward question for those not returning to their preinjury athletics as to "WHY," with a choice of possible factors. There are of course many factors, often not knee related. In regard to knee-related factors, the Svantesson study reports a low follow-up ACL-RSI score of 59.4 and 57.0 (out of a possible 100) in the ACL-MCL and ACL groups.12 The RSI score values are much lower than recommended and reflect issues related to psychological readiness and confidence factors for RTS. In a recent study, Webster et al14 reported on factors that predict a RTS participation in 222 ACL reconstructions evaluated at 1 to 3 years after surgery. A total of 61% returned to their preinjury levels of performance and the study identified a statistical relationship of psychological readiness, greater limb symmetry, higher subjective knee scores, and higher activity levels as associated with a return to performance. Of importance, the only significant RTS predictor was the determination of psychological readiness. The ACL-RSI score (0-100) was reported at 75 on athletes who returned to performance and 64 on athletes who did not return ($P < 0.01$). Also stressed in this study is that return to athletic performance involves stages of return: (1) return to initial sports participation, (2) a return to sports, and (3) return to performance at the chosen sport. The third phase indicates the athlete performs at or above the preinjury level. The 3 RTS phases have different meanings and the ultimate goal is a return to preinjury sports with return of performance indicators. This raises the bar for determining the final RTS and adds on performance criteria determined by either subjective or objective parameters.[14] The final RTS rating usually requires 2 to 3 years for meaningful data, and 1-year data are often too soon for accuracy.

An important question raised in the Svantesson et al[12] study is the prolonged time from injury to ACL reconstruction of 114.6 days and 384 days in the ACL-MCL and ACL groups, respectively. This indicates the patients undergoing ACL reconstruction were usually chronic ACL cases, far removed from participating in high-level competitive athletics. Also reported are associated lateral meniscal tears in 39.2% and medial meniscal tears in 17.5% of the total group. One wonders what the data would show with an earlier intervention of ACL reconstruction and avoiding this extended delay. It would seem the

time delay until ACL reconstruction and the associated meniscal injuries would strongly affect the ability to return to high preinjury athletics.

The current study does follow 2 other publications on ACL reconstruction by these authors using the same registries. In a 2018 publication of patients with ACL reconstruction at 1 year after surgery, positive predictors of return to knee-strenuous sports were male sex, younger age, high preinjury level of athletics, and absence of MCL and meniscal injuries.[10] In a 2019 publication, the authors reported from this database an increased revision rate of ACL reconstructions with a previous nonsurgical-treated MCL injury compared with ACL reconstructions with MCL surgical treatment.[13] The analysis of the registry data of the studies is limited by no classification of medial ligament injuries, and the authors indicate knee medial ligament stability test data would have strengthened the study. There is a lack of data on the types of medial ligament disruption preoperative, or findings at surgery, or at follow-up for ACL function or failure, or medial knee stability. In addition to medial ligament stability tests at surgery, there are objective joint opening gap tests that can be performed at arthroscopy.

The question arises in the ACL-MCL group how many knees underwent subsequent ACL reconstruction with abnormal medial joint opening and residual medial knee instability. An associated medial ligament injury and abnormal medial joint opening may have major adverse effects on the success of ACL reconstruction and clinical results.[1,7,9] This criticism is not just an issue with the above studies. There remains controversy on the treatment of medial ligament injuries in highly functioning athletes and, in particular, with a combined superficial MCL (sMCL), deep MCL, and associated posteromedial corner (PMC) (oblique popliteal ligament complex) disruption that involves consideration for surgical treatment over nonoperative treatment.[11] One of the problems leading to controversy is the lack of objective stability data including stress radiography for classification of the medial ligament injury and medial stability preoperative and postoperative. The results of cadaver ligament cutting studies show an approximate 3 mm of increased medial joint opening with a complete sMCL disruption (30° of knee flexion) that doubles to 6 mm increased opening with added PMC disruptions.[4,6] These objective data are needed for studies arriving at treatment recommendations and includes increases in medial joint opening, external tibial rotation, and internal tibial rotation with PMC disruption.

In conclusion, Svantesson et al12 are congratulated for their continuing efforts in bringing to the orthopaedic community the ongoing results from the largest ACL injury registry that exists. These data provide needed information in many categories related to ACL reconstruction, associated injuries,

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surgery, and postoperative success rates. It is not anticipated that the registry provides all of the information necessary for granular analysis of the data; however, the registry does have the ability to highlight problems needing future Level 1 studies. At the same time, the registry data have limitations where added data and clarity are necessary, which awaits future clinical research. RTS has too many variables that affect the final outcome and may have a limited basis for defining results of surgery, without a granular analysis to separate nonknee factors from knee-related factors. Even so, the goal remains for the return of sports at the chosen level of athletics with return of preinjury performance. It is also recognized that often these ACL injuries and associated concomitant injuries may require modification of sports and athletic expectations to allow a longer term participation in noninjurious athletic activities.

Substance misuse in elite athletes: Early detection, brief intervention and referral to treatment. Clinics in Sports Medicine, Ahead of Print: 1-18, 2023.

KEY POINTS

- Encounters with alcohol, cannabis, and nicotine/tobacco misuse is common.
- Early identification can reduce the negative consequences of misuse and prevent the progression to a substance use disorder.
- The use of screening instruments, negative consequences questionnaires, and urine testing can aid early identification.
- Brief motivational interventions with personalized feedback, social norming, goal setting, contingency management, and significant other involvement can change attitudes, beliefs, and use patterns.
- The simultaneous use of alcohol and marijuana (SAM) among athletes is increasing and is associated with increased adverse effects.

Introduction - Sports medicine physicians and athletic trainers regularly encounter athletes who misuse substances that put them at risk for adverse social, interpersonal, academic, psychological, health and performance effects. The most often encountered substances in collegiate and professional athletes are alcohol (binge drinking), cannabis (marijuana), and nicotine (oral tobacco/nicotine vaping).

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Male athletes typically misuse substances at higher rates than female athletes and male and female athletes from certain sports (eg, lacrosse, ice hockey, swimming, rugby) have the highest rates of use and misuse. Differences in use and misuse rates of some substances in athletes versus non-athletes can begin in high school where, for example, binge drinking rates for athletes is higher than in non-athletes by 9th grade.

Alcohol and other substance use typically begins as recreational or social use in high school and increases from 8th to 12th grades in both athlete and general student populations. In 2022 a national survey demonstrated that past year use rates were significantly higher for 12th graders than 8th graders for alcohol (51.9% vs 15.2%), cannabis (30.7% vs 8.3%), and nicotine vaping (27.3% vs 12%). For some athletes, substance use becomes more regular and heavier, progressing to misuse by the end of high school. Misuse is defined as using alcohol or other drugs in a manner, situation, amount, or frequency that could cause harm to a user or to those around them. Common adverse effects vary somewhat by substance class, but include physical, social, legal, emotional, performance and academic consequences. If substance misuse is identified early, prompt intervention can prevent progression to a substance disorder with or without comorbid mental health symptoms and disorders.

Substances may be used individually or concurrently with an increasing trend for using two together such that their psychogenic and behavioral effects overlap. The simultaneous use of alcohol and marijuana (SAM) and alcohol and tobacco/nicotine is common among college students including student athletes. When two substances are used simultaneously, additive adverse effects such as heavier and more frequent use, driving while impaired, academic and athletic underperformance, and an increased probability of developing comorbid substance use and mental health disorders are likely.

The early identification of athletes who are misusing any of these substances, alone or in combination, is critical to prevent the range of negative consequences and/or progression to a substance use disorder. Behavioral observations, the longitudinal use of screening instruments and/or negative consequences questionnaires, and random urine testing are important mitigation strategies. Given the amount of time that sports medicine physicians and athletic trainers spend with athletes, they are in the best position to pick up early behavioral signs and symptoms of increasing substance use or misuse. The regular use of screening instruments throughout the year, and especially pre-season, should be considered. Short screeners include the three question Alcohol Use Disorders Identification Test-Consumption Questions, the four question Cutting Down, Annoyance by Criticism, Guilty Feeling, and

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Eye Opener Adapted to Include Drugs, and the eight or three question Cannabis Use Disorder Screening Tool Revised or Short Form. For those who screen positive and/or have a positive urine test, a more detailed assessment documenting the current frequency of use, route of administration, and negative consequences using either the Brief Young Adult Alcohol Consequences Questionnaire (B-YAACQ) or the Brief Marijuana Consequences Questionnaire (B-MACQ) should be conducted.

Once alcohol, cannabis, and/or nicotine/tobacco misuse is identified and an in-depth assessment has been conducted, various interventions such as psychoeducation, direct or web-based individualized feedback, goal setting, support system involvement, contingency management and/or brief motivational interventions can be considered. If these interventions are not effective in reducing substance use and/or its adverse effects, then referral to more targeted treatment (eg, outpatient, intensive outpatient, residential, inpatient detoxification) may be necessary.

This article will focus on alcohol, cannabis, and tobacco misuse and explore early detection, brief interventions, and referral to treatment. Additionally, a brief description of the available levels of care for the treatment of severe or intractable substance misuse or a substance use disorder will be presented.

The association between hip/groin pain and hip strength in football players: An exploratory analysis of the FORCe cohort. Journal of Science and Medicine in Sport, Ahead of Print: 1-25, 2023.

Objectives: Explore associations between peak hip strength in football players with hip/groin pain and healthy controls.

Design: Cross-sectional study

Methods: Male and female sub-elite football players (soccer and Australian Football) with hip/groin pain >6-months duration and players without hip/groin pain were recruited across Melbourne and Brisbane, Australia. Demographic information and two questionnaires; the Copenhagen Hip and Groin Outcome Score (HAGOS) and the International Hip Outcome Tool 33 (iHOT) were collected. Hand-held dynamometry was used to measure isometric hip strength for flexion, extension, abduction, adduction,

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internal rotation, and external rotation. Linear mixed effects models were used to compare strength measures between groups.

Results: 190 football players with hip/groin pain (mean±SD age, 27.8±6.3 years) and 64 controls (age, 27.3±5.6 years) were included in this study. Of these, 291 symptomatic limbs and 128 control limbs were used for analyses. Symptomatic players had lower peak hip adduction (adjusted mean difference= -0.18: 95%CI -0.27 to -0.08, P=0.001), external rotation (-0.06: 95%CI -0.09 to -0.02, P=0.003), and internal rotation strength (-0.06: 95%CI -0.10 to -0.03, P=0.001) compared to controls. A sport-specific interaction was observed for hip abduction strength. When separated by football code, abduction strength was lower in symptomatic Australian football players compared to their same sport peers (-0.20: 95%CI -0.33 to -0.06, P=0.004), but not in symptomatic soccer players (-0.05: 95%CI -0.15 to 0.06, P=0.382).

Conclusions: Hip adduction, internal rotation, and external rotation strength appear lower in football players with hip/groin pain independent of sex and football code. Hip abduction strength was lower in symptomatic Australian football players but not in soccer players.

The role of strength-related factors on psychological readiness for return to sport following anterior cruciate ligament (ACL) reconstruction. Healthcare, 11: 2787, 2023.

Abstract - Psychological readiness following anterior cruciate ligament reconstruction (ACLR) correlates with different return to sport outcomes. However, the relationship between strength and power and psychological readiness remains unexplored. The aim of this study was to investigate the relationship between anterior cruciate ligament return to sport after injury (ACL-RSI) scores and various hamstrings and quadriceps strength and power variables. Twelve participants (20.7 ± 2.5 years old; 174.2 ± 7.5 cm; 70.2 ± 8.5 kg; 18.2 ± 8.3% of body fat) who had an ACLR nine months or more before the study completed the ACL-RSI questionnaire and isokinetic strength testing of the hamstrings and quadriceps ($60^{\circ} \cdot s^{-1}$ and $180^{\circ} \cdot s^{-1}$). Based on ACL-RSI scores, they were divided into “cases” and “controls”, deemed not psychologically ready and psychologically ready to return to previous sport performance (PILOS), respectively. The main findings are that quadriceps’ and hamstrings’ rate of torque

development (RTD) and time since surgery were determinants of psychological readiness following ACLR. Furthermore, compared to controls, cases showed significantly lower quadriceps torque at angles close to full knee extension (40 deg and 30 deg from extension). They also showed lower RTD than controls, but no difference in peak torque. These results suggest that physiotherapists should facilitate athletes' return to sport (RTS) by focusing on the restoration of RTD and strength at angles close to full knee extension.

Utilizing extracorporeal shockwave therapy for in-season athletes. Healthcare, 11: 1006, 2023.

Overview - An athlete's health and availability to train and compete at an optimal performance level is a growing focus for professional sports organizations. Back-to-back competitions with limited recovery time, along with travelling across multiple time zones, are inherent challenges. Studies suggest that a higher rate of injuries are sustained during the pre-season and during competitions. These injuries include, but are not limited to, muscle strains [3], tendinopathies [4], and injuries to bone. Athletic injuries can be challenging to treat in-season, with unpredictable healing times following treatment interventions. Surgical management is typically reserved for off-season athletes, as recovery commonly requires a substantial rehabilitation period of 6 to 9 months post-operatively [4]. Identifying methods to treat injuries that exert positive clinical effects within 3 months would be desirable for in-season athletes. Emerging research suggests that extracorporeal shockwave therapy (ESWT) may represent an effective treatment to address sports-related injuries for in-season athletes and accelerate return to play.

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